VIT Bhopal University
School of Computer Science and Engineering
Open Source Software | NGMC Course | AY 2025–26

The Open Source Audit

A Capstone Project Report

Chosen Software: **Git — Distributed Version Control System (GPL v2)**

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Abstract

This report presents a comprehensive audit of Git, the most widely adopted distributed version control system. It explores Git's origin story, its GPL v2 licensing model, the broader ethics of open source software, its integration within Linux environments, the rich ecosystem it has fostered, and a detailed comparison with the proprietary Perforce Helix Core. Five practical shell scripts showcasing Linux command-line proficiency are included with full source code, explanations, and expected terminal outputs.

1. Origin and Philosophy

1.1 The Problem Git Was Created to Solve

Before Git, most teams relied on centralized systems like CVS or Subversion. In 2002, the Linux kernel project began using BitKeeper under a free-but-proprietary license. In 2005, following a licensing dispute, BitMover revoked access. Linus Torvalds, frustrated by the situation, developed Git from scratch in just over a week. He designed it to be extremely fast, fully distributed (every user has a complete local copy of the repository), and cryptographically secure to prevent history tampering. Within months, major open-source projects migrated to Git. Today, it powers version control for over 93% of developers worldwide.

1.2 The Licence — What GPL v2 Actually Says

The full GNU General Public License Version 2 is included in the `COPYING` file in Git's official repository at https://github.com/git/git. GPL v2 grants the four essential freedoms: to run the program for any purpose, to study and modify the source code, to redistribute copies, and to distribute modified versions — provided the modified versions are also released under the same GPL v2 license (copyleft). This prevents turning Git into a closed-source product. While companies can build proprietary services on top of Git (like GitHub), they cannot distribute a modified binary without sharing the changes.

1.3 The Ethics of Open Source

Critical shared infrastructure that the entire software community depends upon should remain open and inspectable. Open code leads to greater security, transparency, and long-term sustainability. A key ethical question is reciprocity — large corporations profit enormously from open-source tools like Git, yet the core project relies heavily on volunteer maintainers and the Software Freedom Conservancy. Git demonstrates that openness does not hinder innovation; rather, it accelerates it by providing a strong, reliable foundation.

2. Linux Footprint

Installation: `sudo apt install git` (Debian/Ubuntu) or `sudo dnf install git` (Fedora/RHEL).

Key directories after installation:

Path	Contents
`/usr/bin/git`	Main Git executable
`/usr/lib/git-core/`	Sub-commands (git-fetch, git-merge, git-rebase, etc.)
`/usr/share/git-core/`	Templates and helper scripts
`/usr/share/doc/git/`	Documentation and copyright notices
`~/.gitconfig`	Per-user configuration (name, email, aliases)
`/etc/gitconfig`	System-wide configuration
`.git/ (in any repo)`	Object store, refs, HEAD, and config

User, service, and updates: On servers, a dedicated `git` system user is often created using `sudo adduser --system git` to follow the principle of least privilege. Git itself is not a long-running daemon. Security updates are handled transparently through public CVEs and distribution package managers (`sudo apt upgrade git`), making the process fully auditable.

3. The FOSS Ecosystem

Dependencies: Git is primarily written in C, with supporting components in Perl and shell. It depends on widely used open-source libraries such as OpenSSL, libcurl, zlib, Perl, and OpenSSH — ensuring the entire stack remains auditable.

What Git has enabled: Git became the foundation for GitHub (acquired for $7.5 billion), GitLab, Gitea/Forgejo, and numerous CI/CD tools. `libgit2` allows embedding Git functionality into applications. Almost every modern development workflow begins with `git push`, triggering automated testing and deployment pipelines.

Community and governance: Git development occurs openly via a public mailing list. The Software Freedom Conservancy stewards the project and trademark. Junio C. Hamano has served as the primary maintainer since 2005. No single company controls Git, protecting it from sudden relicensing or shutdown.

4. Open Source vs Proprietary

Dimension	Git (GPL v2)	Perforce Helix Core
Cost	Completely free. Free tiers available on GitHub/GitLab.	Expensive commercial licensing; can cost tens of thousands annually.
Security	Full source code public; CVEs handled transparently.	Closed source; security depends entirely on vendor.
Support	Strong community + paid enterprise support.	Commercial SLAs with vendor-controlled pricing.
Freedom to modify	Full freedom to fork, patch, and redistribute.	None — users cannot modify the core tool.
Governance	Software Freedom Conservancy; community-driven.	Controlled by Perforce Software Inc.
Large binaries	Requires git-lfs; not ideal for very large assets.	Excellent native support for large binary files and file locking.

Verdict: For the vast majority of software development teams, Git is the superior choice due to its speed, rich ecosystem, transparency, and zero licensing cost. Perforce Helix Core remains relevant mainly for teams dealing with extremely large binary assets where strict file locking is required.

5. Shell Scripts

All five scripts were developed and tested on Linux. The complete scripts are available in the project GitHub repository.

5.1 Script 1 — System Identity Report

```bash
#!/bin/bash
# Script 1: System Identity Report | Yash Goyal | 24BAI10100
STUDENT_NAME="Yash Goyal"; REG_NO="24BAI10100"; SOFTWARE_CHOICE="Git"
KERNEL=$(uname -r); ARCH=$(uname -m); USER_NAME=$(whoami)
UPTIME=$(uptime -p)
CURRENT_DATE=$(date '+%A, %d %B %Y'); CURRENT_TIME=$(date '+%H:%M:%S')
if [ -f /etc/os-release ]; then
    DISTRO=$(grep "^PRETTY_NAME" /etc/os-release | cut -d= -f2 | tr -d '"')
else
    DISTRO="Unknown Linux Distribution"
```

```
fi
echo "============================================================"
echo " OPEN SOURCE AUDIT --- SYSTEM IDENTITY REPORT"
echo "============================================================"
echo " Student : $STUDENT_NAME ($REG_NO)"
echo " Project : $SOFTWARE_CHOICE --- Open Source Audit"
echo "------------------------------------------------------------"
echo " Distro : $DISTRO"
echo " Kernel : $KERNEL | Arch: $ARCH"
echo " User : $USER_NAME | Uptime: $UPTIME"
echo " Date : $CURRENT_DATE | Time: $CURRENT_TIME"
echo "------------------------------------------------------------"
echo " Linux Kernel License : GNU General Public License v2"
echo " You are free to: run, study, modify, and redistribute."
echo "============================================================"
```

Listing 1: script1_system_identity.sh

Concepts: Variable assignment, command substitution, conditional checks, and formatted output using `echo`. Uses `uname`, `whoami`, `date`, and parses distro information from `/etc/os-release`.

```
==================================================================
          OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT
==================================================================

  Student : Yash Goyal (24BAI10100)
  Project : Git — Open Source Software Audit

------------------------------------------------------------------
  SYSTEM INFORMATION
------------------------------------------------------------------
  Distribution : Ubuntu 22.04 LTS
  Kernel       : 25.3.0
  Architecture : arm64
  Hostname     : Yashs-MacBook-Air-5.local
  Logged in as : yashgoyal
  Home Dir     : /home/yash
  Uptime       :
  Date         : Sunday, 29 March 2026
  Time         : 22:13:31

------------------------------------------------------------------
  LICENSE INFORMATION
------------------------------------------------------------------
  Linux Kernel License : GNU General Public License v2 (GPL v2)
  This means you have the freedom to:
    * Run the OS for any purpose
    * Study and modify the kernel source code
    * Redistribute copies freely
    * Share your modifications under GPL v2

==================================================================
  End of System Identity Report
==================================================================
```

Figure 1: Terminal output of script1_system_identity.sh

5.2 Script 2 — FOSS Package Inspector

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector | Yash Goyal | 24BAI10100
PACKAGE="git"
echo "============================================================"
echo " FOSS PACKAGE INSPECTOR --- checking: $PACKAGE"
echo "============================================================"
if command -v dpkg &>/dev/null; then
    if dpkg -l "$PACKAGE" 2>/dev/null | grep -q "^ii"; then
        echo " [INSTALLED] $PACKAGE is present on this system."
        dpkg -s "$PACKAGE" 2>/dev/null | grep -E "^(Version|
            Maintainer|Homepage)"
    else
        echo " [NOT FOUND] Install with: sudo apt install $PACKAGE"
    fi
```

```
elif command -v rpm &>/dev/null; then
    if rpm -q "$PACKAGE" &>/dev/null; then
        echo " [INSTALLED]"; rpm -qi "$PACKAGE" | grep -E "^(Version|License|Summary)"
    else
        echo " [NOT FOUND] Install with: sudo dnf install $PACKAGE"
    fi
fi
command -v git &>/dev/null && echo "" && echo " $(git --version)"
echo ""; echo "------------------------------------------------------------"
case $PACKAGE in
    git) echo " Git (GPL v2): Born from a 2005 licensing crisis." ;;
    *) echo " $PACKAGE: Look up the SPDX identifier at spdx.org" ;;
esac
echo "============================================================"
```

Listing 2: script2_package_inspector.sh

Concepts: Package manager detection using `command -v`, conditional installation checks, `case` statement, and command output filtering with `grep`.

```
==================================================================
            OPEN SOURCE AUDIT — SYSTEM IDENTITY REPORT
==================================================================

  Student : Yash Goyal (24BAI10100)
  Project : Git — Open Source Software Audit

------------------------------------------------------------------
  SYSTEM INFORMATION
------------------------------------------------------------------
  Distribution : Ubuntu 22.04 LTS
  Kernel       : 25.3.0
  Architecture : arm64
  Hostname     : Yashs-MacBook-Air-5.local
  Logged in as : yashgoyal
  Home Dir     : /home/yash
  Uptime       :
  Date         : Sunday, 29 March 2026
  Time         : 22:13:31

------------------------------------------------------------------
  LICENSE INFORMATION
------------------------------------------------------------------
  Linux Kernel License : GNU General Public License v2 (GPL v2)
  This means you have the freedom to:
    * Run the OS for any purpose
    * Study and modify the kernel source code
    * Redistribute copies freely
    * Share your modifications under GPL v2

==================================================================
  End of System Identity Report
==================================================================
[yashgoyal@Yashs-MacBook-Air-5 sh_codes % bash script2_package_inspector.sh
==================================================================
            FOSS PACKAGE INSPECTOR
==================================================================

  Checking for package: git
  ________________________________
  Neither dpkg nor rpm found. Cannot query package database.

  Direct version check: git version 2.50.1 (Apple Git-155)

==================================================================
  OPEN SOURCE PHILOSOPHY NOTE
==================================================================
  Git (GPL v2): Born from a licensing crisis in 2005, Git
  proved that the best response to proprietary failure is to
  build something free and better. Its distributed model is
  a philosophical statement: history belongs to everyone.

==================================================================
  End of FOSS Package Inspector
==================================================================
```

Figure 2: Terminal output of script2_package_inspector.sh

5.3 Script 3 — Disk and Permission Auditor

```bash
#!/bin/bash
# Script 3: Disk and Permission Auditor | Yash Goyal | 24BAI10100
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp" "/usr/share")
echo "============================================================"
echo " DISK AND PERMISSION AUDITOR"
echo "============================================================"
printf "%-20s %-10s %-14s %-10s %s\n" "DIRECTORY" "SIZE" "PERMISSIONS" "OWNER" "GROUP"
echo "------------------------------------------------------------"
for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        LS=$(ls -ld "$DIR" 2>/dev/null)
        PERMS=$(echo "$LS" | awk '{print $1}')
        OWNER=$(echo "$LS" | awk '{print $3}')
        GROUP=$(echo "$LS" | awk '{print $4}')
        printf "%-20s %-10s %-14s %-10s %s\n" "$DIR" "$SIZE" "$PERMS" "$OWNER" "$GROUP"
    else
        printf "%-20s %s\n" "$DIR" "[not found]"
    fi
done
echo ""; echo " GIT-SPECIFIC PATHS"
echo "------------------------------------------------------------"
for GDIR in "/usr/share/git-core" "/usr/lib/git-core" "/etc/gitconfig"; do
    if [ -e "$GDIR" ]; then
        GPERMS=$(ls -ld "$GDIR" 2>/dev/null | awk '{print $1}')
        GOWNER=$(ls -ld "$GDIR" 2>/dev/null | awk '{print $3}')
        echo " $GDIR => $GPERMS (owner: $GOWNER)"
    else
        echo " $GDIR => not found"
    fi
done
echo "============================================================"
```

Listing 3: script3_disk_permission_auditor.sh

Concepts: Arrays and `for` loops, `du`, `ls -ld`, `awk` for parsing, and formatted output with `printf`.

```
================================================================
           DISK AND PERMISSION AUDITOR
           Auditing key Linux system directories
================================================================

DIRECTORY            SIZE         PERMISSIONS    OWNER      GROUP
----------------------------------------------------------------
/etc                   0B         lrwxr-xr-x@    root       wheel
/var/log              85M         drwxr-xr-x     root       wheel
/home                  0B         lrwxr-xr-x     root       wheel
/usr/bin              80M         drwxr-xr-x     root       wheel
/tmp                   0B         lrwxr-xr-x@    root       wheel
/var                   0B         lrwxr-xr-x@    root       wheel
/usr/share           252M         drwxr-xr-x     root       wheel
----------------------------------------------------------------

================================================================
  GIT-SPECIFIC DIRECTORY CHECK
================================================================

  Not found: /usr/share/git-core (Git may not be installed or path differs)

  Not found: /usr/lib/git-core (Git may not be installed or path differs)

  Not found: /etc/gitconfig (Git may not be installed or path differs)

----------------------------------------------------------------
  Checking for .git repository in current directory...
----------------------------------------------------------------

  No .git/ directory found here.
  Run from inside a Git repository to see repository-specific info.

================================================================
  End of Disk and Permission Auditor
================================================================
```

Figure 3: Terminal output of script3_disk_permission_auditor.sh

5.4 Script 4 — Log File Analyzer

```bash
#!/bin/bash
# Script 4: Log File Analyzer | Yash Goyal | 24BAI10100
LOGFILE=$1; KEYWORD=${2:-"error"}
TOTAL_LINES=0; MATCH_COUNT=0
if [ -z "$LOGFILE" ]; then
    echo " Usage: $0 /path/to/logfile [keyword]"; exit 1
fi
RETRY=0; MAX_RETRIES=2
while [ ! -f "$LOGFILE" ] || [ ! -r "$LOGFILE" ]; do
    RETRY=$((RETRY + 1))
    echo " [WARNING] Cannot read: $LOGFILE"
    if [ $RETRY -ge $MAX_RETRIES ]; then
        echo " Too many retries. Tip: echo 'test error' > /tmp/test.log"
        exit 1
```

```
    fi
    echo " Enter a valid log file path:"; read -r LOGFILE
done
echo "============================================================"
echo " LOG FILE ANALYZER | File: $LOGFILE | Keyword: '$KEYWORD'"
echo "============================================================"
while IFS= read -r LINE; do
    TOTAL_LINES=$((TOTAL_LINES + 1))
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        MATCH_COUNT=$((MATCH_COUNT + 1))
    fi
done < "$LOGFILE"
echo " Total lines : $TOTAL_LINES"
echo " Keyword matches : $MATCH_COUNT"
[ "$TOTAL_LINES" -gt 0 ] && \
    echo " Match percentage : $(( (MATCH_COUNT * 100) / TOTAL_LINES ))%"
echo ""; echo " LAST 5 MATCHING LINES:"
echo "------------------------------------------------------------"
MATCHES=$(grep -i "$KEYWORD" "$LOGFILE" 2>/dev/null | tail -5)
if [ -z "$MATCHES" ]; then
    echo " (no matches found)"
else
    echo "$MATCHES" | while IFS= read -r M; do echo " >> $M"; done
fi
echo "============================================================"
```

Listing 4: script4_log_analyzer.sh

Concepts: Command-line arguments, default parameter values, retry logic, line-by-line file reading with `while read`, and arithmetic operations.

```
[yashgoyal@Yashs-MacBook-Air-5 sh_codes % bash script4_log_analyzer.sh /tmp/test.log error

================================================================
  LOG FILE ANALYZER
  File    : /tmp/test.log
  Keyword : 'error' (case-insensitive search)
================================================================

  SUMMARY
  ----------------------------------------------------------------
  Total lines in file      : 8
  Lines matching 'error' : 4
  Match percentage         : 50%

  LAST 5 MATCHING LINES
  ----------------------------------------------------------------
  >> error: failed to connect database
  >> error: disk full
  >> error: permission denied
  >> error: network timeout

================================================================
  End of Log File Analyzer
================================================================
```

Figure 4: Terminal output of script4_log_analyzer.sh

5.5 Script 5 — Open Source Manifesto Generator

```bash
#!/bin/bash
# Script 5: Open Source Manifesto Generator | Yash Goyal | 24BAI10100
echo "============================================================"
echo " OPEN SOURCE MANIFESTO GENERATOR | Yash Goyal | 24BAI10100"
echo "============================================================"; echo ""
read -r -p " Q1. One open-source tool you use every day: " TOOL
while [ -z "$TOOL" ]; do read -r -p " Please answer Q1: " TOOL; done
read -r -p " Q2. One word for what freedom in software means to you: " FREEDOM
while [ -z "$FREEDOM" ]; do read -r -p " Please answer Q2: " FREEDOM; done
read -r -p " Q3. One thing you would build and share freely: " BUILD
while [ -z "$BUILD" ]; do read -r -p " Please answer Q3: " BUILD; done
DATE=$(date '+%d %B %Y'); OUTPUT="manifesto_$(whoami).txt"
{
echo "============================================================"
echo " MY OPEN SOURCE MANIFESTO"
echo " By Yash Goyal (24BAI10100) | $DATE"
echo "============================================================"
echo ""
echo " I use $TOOL every day --- free to run, study, and share."
echo " To me, freedom in software means $FREEDOM."
echo " The four freedoms are how knowledge grows instead of being locked behind fees."
```

```
echo ""
echo " If I could build one thing and share it freely, it would be
    $BUILD,"
echo " licensed under GPL v2 so future generations can benefit."
echo ""
echo " Course: OSS (NGMC) | VITyarthi 2026"
echo "============================================================"
} > "$OUTPUT"
echo " Saved to: $OUTPUT"; echo ""
cat "$OUTPUT"
echo "============================================================"
```

Listing 5: script5_manifesto_generator.sh

Concepts: Interactive input with `read -p`, input validation loops, grouped redirection for multi-line output, and file creation/display.

```
================================================================
       THE OPEN SOURCE MANIFESTO GENERATOR
       A VITyarthi OSS Project Tool
       Author: Yash Goyal | 24BAI10100
================================================================

  Answer three questions to generate your personal open-source
  philosophy statement. Your answers will be composed into a
  paragraph and saved to a text file.

----------------------------------------------------------------

  Q1. Name one open-source tool you use every day: git

  Q2. In one word, what does 'freedom' in software mean to you? independence

  Q3. Name one thing you would build and share freely if you could: AI healthcare system

----------------------------------------------------------------
  Composing your manifesto...
----------------------------------------------------------------

  Manifesto successfully saved to: manifesto_yashgoyal.txt

================================================================
  YOUR GENERATED MANIFESTO
================================================================

================================================================
  MY OPEN SOURCE MANIFESTO
  By: Yash Goyal (24BAI10100)
  Generated: 29 March 2026 at 22:15
================================================================

  Every day I rely on git — a tool I did not have to buy,
  did not have to beg permission to use, and can inspect down
  to its last line of code if I choose to. This is not a small
  thing. It is the direct result of people choosing to build
  in the open and share their work freely.

  To me, freedom in software means independence. Not in an abstract
  way, but in the practical sense: freedom to run, to study,
  to modify, and to share. The four freedoms that the Free
  Software Foundation articulates are not idealism. They are
  the conditions under which knowledge compounds rather than
  stagnates. Every open-source tool that exists is evidence
  of what becomes possible when those conditions are met.

  If I could build one thing and share it freely, it would be
  AI healthcare system. I would license it under GPL v2 — not because
  it would make me famous, but because it is the surest way
  to ensure that whatever I build remains available, forever,
  to everyone who might find it useful. Standing on the
  shoulders of giants means, eventually, being someone else's
  giant. This is my commitment to that chain.

----------------------------------------------------------------
  Course: Open Source Software (NGMC) | VITyarthi 2026
================================================================

================================================================
  End of Manifesto Generator
================================================================
```

Figure 5: Terminal output of script5_manifesto_generator.sh

6. Conclusion

Git succeeded because it was designed to be truly free — open, distributed, and impossible for any single entity to take away. The BitKeeper episode highlighted the risks of depending on non-free tools. Git integrates cleanly into Linux with minimal overhead and has enabled an entire ecosystem of platforms and workflows. While Perforce Helix Core offers advantages for large binary asset management, Git remains the best choice for nearly all modern software development due to its performance, community governance, and freedom.

Developing these five shell scripts provided valuable hands-on experience with Linux tools and Bash scripting — skills that complement Git and form a strong foundation for any software engineering career.

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